SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

ICT 133
Structured Programming

Tutor-Marked Assignment

July 2025 Presentation

TUTOR-MARKED ASSIGNMENT (TMA)

This assignment is worth **24%** of the final mark for **ICT133, Structured Programming.**

The cut-off date for this assignment is **Sunday, 12 October 2025, 2355 hours**.

Note to Students:

All TMAs must be submitted electronically through Canvas. No partial submission of TMA will be accepted unless otherwise specified.

Assignment Requirements:

- Do **NOT** define classes for this TMA.
- **Unless specified in the question, you CANNOT use packages not covered in this module, e.g., re, collections, numpy, pandas etc.**
- All functions must be documented. Provide sufficient comments to your code and ensure that your program adheres to good programming practices such as not using global variables.
- **Do not use exit() function.**
- Failing to do so can incur a penalty of as much as 50% of the marks allotted.

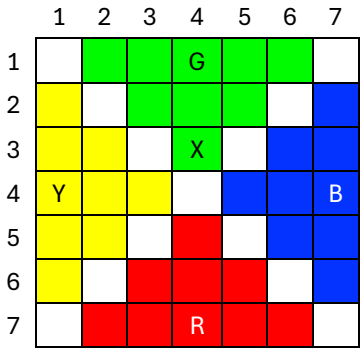
Submission Details:

- Use the template word document provided - **SUSS_PI_No-FullName_133TMA.docx.** Rename the file with your SUSS PI and full name join with "_133TMA" e.g., "**SUSS_PI-TomTanKinMeng_133TMA.docx**" (without the quotes).
- Include the following particulars on the first page of the word document, on separate lines: **Course Code, SUSS PI No., Your Name, Tutorial Group and Submission Date.**
- Copy and paste the source code of each program you write in **text** format. Submit screenshots for **only** output of your program, where applicable.
- If you submit the source code as a screenshot, your code will not be marked. That is, **screenshot code will be awarded zero mark.**
- Submit your solution in the form of a **single MS Word document. Do NOT submit as a pdf document.** You will be penalised if you fail to submit a word document.
- The word document must be submitted to your respective T group. Besides this, you are required to upload your source code to Vocareum. Do a **print screen** of the Vocareum submission and insert this screenshot into your word document.

Question 1 (20 marks)

This question covers materials in Seminar 1 and 2. Use and express selection structure for this question. Functions, repetition, or collections are not necessary for this question.

(a) Write a program to find the colour of square **X** on a board with colour patterns. Your program will ask a user to input the size of the board, which can be any odd integer from 3 to 9 inclusive. Imagine 2 diagonal lines cutting across the board with no colours. The top triangle with be **G**reen, bottom **R**ed, left **Y**ellow and right **B**lue. A size 7 board example is shown below.



The program will ask the user for the row and column numbers of square **X**. Column numbers start from 1 and run from left to right. Row numbers start from 1 and run from top to bottom. If **X** falls on the diagonals, the program shall print 'No colour', else it shall print the colour of square **X**. The program shall do error checking on size, row, column numbers and not ask the user for unnecessary inputs if the previous input already failed the check. Sample runs are given below. Submit and paste screenshots of at least FOUR program executions, with different inputs.

<u>Sample 1:</u>
```
Enter size of board (3-9, odd): 8
Invalid board size! Please enter an odd integer between 3 and 9
```

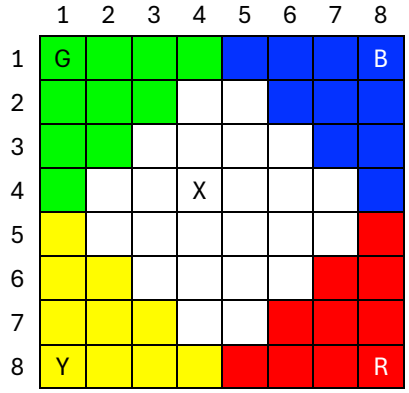
<u>Sample 2:</u>
```
Enter size of board (3-9, odd): 3
Enter the row number: 2
Enter the column number: 1
Colour is Yellow
```

<u>Sample 3:</u>
```
Enter size of board (3-9, odd): 7
Enter the row number: 3
Enter the column number: 4
Colour is Green
```

<u>Sample 4:</u>
```
Enter size of board (3-9, odd): 9
Enter the row number: 7
Enter the column number: 3
No colour
```

(10 marks)

(b) The pattern of the board has been changed to the type below. Write a program to ask a user to input the size of the board, which now must be an even integer from 4 to 10 inclusive. The program will ask the user for the row and column numbers of square **X** and print out the colour of square **X** or 'No colour' if it falls within the central white space. Submit and paste screenshots of at least FOUR program executions, with different inputs.



<u>Sample 1:</u>

```
Enter size of board (4-10, even): 4
Enter the row number: 3
Enter the column number: 1
Colour is Yellow
```

<u>Sample 2:</u>

```
Enter size of board (4-10, even): 6
Enter the row number: 3
Enter the column number: 1
Colour is Green
```

<u>Sample 3:</u>

```
Enter size of board (4-10, even): 8
Enter the row number: 6
Enter the column number: 7
Colour is Red
```

<u>Sample 4:</u>

```
Enter size of board (4-10, even): 10
Enter the row number: 7
Enter the column number: 3
No colour
```

(10 marks)

Question 2 (30 marks)

This question covers materials up to Seminar 3. Make use of functions, selection and repetition structures. NO data structures like sets, lists or dictionary should be used for this question. The eval() function should not be used. Keep the program modular by defining other functions if necessary.

(a) Make changes to the answer to 1(b) above and convert it into a function that receives input via its input parameters and return the result via its return values. Write a function getColour(size, row, col) which takes in 3 integer parameters containing the size of the board, row and column numbers of square **X**. The function returns a single character value that represents the colour of square **X**: 'G', 'B', 'R', 'Y', '-' for Green, Blue, Red, Yellow and no colour respectively. If an error occurred, return an error message string that starts with 'E'.

For example,

- `getColour(8,5,7)` returns '-'
- `getColour(8,6,7)` returns 'R'

(10 marks)

(b) Making use of the function defined in part (a), write a program that will ask a user to input the size of the board, which must be an even integer from 4 to 10 inclusive, and prints out the board using characters to represent the colours. Sample runs are given below. Submit and paste screenshots of at least FOUR program executions, with different inputs.

<u>Sample 1:</u>

```
Enter size of board (4-10, even): 3
Invalid board size! Please enter an even integer between 4 and 10
```

<u>Sample 2:</u>

```
Enter size of board (4-10, even): 6
GGGBBB
GG--BB
G----B
Y----R
YY--RR
YYYRRR
```

<u>Sample 3:</u>

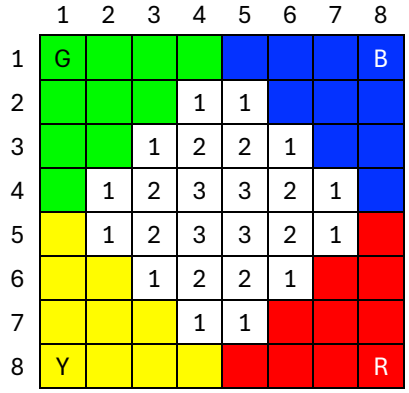
```
Enter size of board (4-10, even): 8
GGGGBBBB
GGG--BBB
GG----BB
G------B
Y------R
YY----RR
YYY--RRR
YYYYRRRR
```

Sample 4:

```
Enter size of board (4-10, even): 10
GGGGGBBBBB
GGGG--BBBB
GGG----BBB
GG------BB
G--------B
Y--------R
YY------RR
YYY----RRR
YYYY--RRRR
YYYYYRRRRR
```

(10 marks)

(c) Enhance the above to print out a board where rings of numbers cover the region with no colours, starting with 1 at the outermost ring and increasing by 1 as it moves towards the centre. Sample runs are given below. NO data structures like sets, lists or dictionaries should be used. Submit and paste screenshots of at least FOUR program executions, with different inputs.



Sample 1:

```
Enter size of board (4-10, even): 4
GGBB
G11B
Y11R
YYRR
```

Sample 2:

```
Enter size of board (4-10, even): 6
GGGBBB
GG11BB
G1221B
Y1221R
YY11RR
YYYRRR
```

Sample 3:

```
Enter size of board (4-10, even): 8
GGGGBBBB
GGG11BBB
GG1221BB
G123321B
Y123321R
YY1221RR
YYY11RRR
YYYYRRRR
```

Sample 4:

```
Enter size of board (4-10, even): 10
GGGGGBBBBB
GGGG11BBBB
GGG1221BBB
GG123321BB
G12344321B
Y12344321R
YY123321RR
YYY1221RRR
YYYY11RRRR
YYYYYRRRRR
```

(10 marks)

Question 3 (15 marks)

This question covers materials up to seminar 4. Employ structure programming and use of functions to make the program modular.

Write a multi-player game where everyone starts with a score of 501 points, with the aim of deducting it to 0 points within the least number of turns.

- The program first asks for a list of unique player names, minimum 2 maximum 5 players.
- The program will shuffle the list of players, with each player taking turns to play.
- For each turn, a player can have up to 3 rounds.
- For each round, the player will first choose a number between 1 to 20 inclusive or 25 to serve as the hit points.
- The program will then randomly choose a multiplier of 1 or 2 if the chosen number is 25, or a multiplier of 1,2 or 3 if the chosen number is 1 to 20.
- The product of these 2 numbers will be deducted from the player's score.
- The player would win if the player's score ended up exactly zero provided the multiplier was 2. Display the number of turns he took and the game ends. For example,
    - Player's current score is 16
    - Player chose 8 as the hit points
    - Program randomly chose a multiplier of 2
    - Player's new score will be 16 – 8*2 = 0 points, which is a win

- The player would go bust if the deduction resulted in a negative score, or a zero score when the multiplier was not 2, or a score of 1 which cannot be deducted to zero with a multiplier of 2 in the next round. The player's score would then revert to his initial score at the start of his turn and the turn will pass to the next player. For example,
    - Player's current score is 16
    - Player chose 8 as the hit points
    - Program randomly chose a multiplier of 3
    - If deducted, the score will be negative, i.e. 16 – 8*3 = -8, which is a bust
- If the player did not win and did not go bust, he will play 3 rounds before the turn passes to the next player.

A sample run is given below. Submit and paste screenshots to show a sample run of your game.

```
Enter player 1's name (empty to end): alan
Enter player 2's name (empty to end):
Please enter at least 2 players
Enter player 2's name (empty to end): Alan
Player names must be unique
Enter player 2's name (empty to end): tom
Enter player 3's name (empty to end): john
Enter player 4's name (empty to end):
Tom's turn 1, current score is 501 points
Tom's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, Tom's new score is 481 points
Tom's round 2, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, Tom's new score is 461 points
Tom's round 3, choose a number from 1 to 20 inclusive, or 25: 20
It's a double, total 2x20=40 points, Tom's new score is 421 points
Next player...
Alan's turn 1, current score is 501 points
Alan's round 1, choose a number from 1 to 20 inclusive, or 25: 25
It's a double, total 2x25=50 points, Alan's new score is 451 points
Alan's round 2, choose a number from 1 to 20 inclusive, or 25: 25
It's a double, total 2x25=50 points, Alan's new score is 401 points
Alan's round 3, choose a number from 1 to 20 inclusive, or 25: 25
It's a single, total 1x25=25 points, Alan's new score is 376 points
Next player...
John's turn 1, current score is 501 points
John's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a triple, total 3x20=60 points, John's new score is 441 points
John's round 2, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, John's new score is 421 points
John's round 3, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, John's new score is 401 points
Next player...
Tom's turn 2, current score is 421 points
Tom's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, Tom's new score is 401 points
Tom's round 2, choose a number from 1 to 20 inclusive, or 25: 20
It's a double, total 2x20=40 points, Tom's new score is 361 points
Tom's round 3, choose a number from 1 to 20 inclusive, or 25: 20
```

```
It's a double, total 2x20=40 points, Tom's new score is 321 points
Next player...
Alan's turn 2, current score is 376 points
Alan's round 1, choose a number from 1 to 20 inclusive, or 25: 25
It's a double, total 2x25=50 points, Alan's new score is 326 points
Alan's round 2, choose a number from 1 to 20 inclusive, or 25: 25
It's a single, total 1x25=25 points, Alan's new score is 301 points
Alan's round 3, choose a number from 1 to 20 inclusive, or 25: 25
It's a single, total 1x25=25 points, Alan's new score is 276 points
Next player...
John's turn 2, current score is 401 points
John's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a triple, total 3x20=60 points, John's new score is 341 points
John's round 2, choose a number from 1 to 20 inclusive, or 25: 20
It's a triple, total 3x20=60 points, John's new score is 281 points
John's round 3, choose a number from 1 to 20 inclusive, or 25: 20
It's a triple, total 3x20=60 points, John's new score is 221 points
Next player...
Tom's turn 3, current score is 321 points
Tom's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a double, total 2x20=40 points, Tom's new score is 281 points
Tom's round 2, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, Tom's new score is 261 points
Tom's round 3, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, Tom's new score is 241 points
Next player...
Alan's turn 3, current score is 276 points
Alan's round 1, choose a number from 1 to 20 inclusive, or 25: 25
It's a double, total 2x25=50 points, Alan's new score is 226 points
Alan's round 2, choose a number from 1 to 20 inclusive, or 25: 25
It's a double, total 2x25=50 points, Alan's new score is 176 points
Alan's round 3, choose a number from 1 to 20 inclusive, or 25: 25
It's a single, total 1x25=25 points, Alan's new score is 151 points
Next player...
John's turn 3, current score is 221 points
John's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a double, total 2x20=40 points, John's new score is 181 points
John's round 2, choose a number from 1 to 20 inclusive, or 25: 20
It's a double, total 2x20=40 points, John's new score is 141 points
John's round 3, choose a number from 1 to 20 inclusive, or 25: 20
It's a triple, total 3x20=60 points, John's new score is 81 points
Next player...
Tom's turn 4, current score is 241 points
Tom's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a double, total 2x20=40 points, Tom's new score is 201 points
Tom's round 2, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, Tom's new score is 181 points
Tom's round 3, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, Tom's new score is 161 points
Next player...
Alan's turn 4, current score is 151 points
Alan's round 1, choose a number from 1 to 20 inclusive, or 25: 25
It's a double, total 2x25=50 points, Alan's new score is 101 points
Alan's round 2, choose a number from 1 to 20 inclusive, or 25: 25
```

```
It's a single, total 1x25=25 points, Alan's new score is 76 points
Alan's round 3, choose a number from 1 to 20 inclusive, or 25: 25
It's a double, total 2x25=50 points, Alan's new score is 26 points
Next player...
John's turn 4, current score is 81 points
John's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a single, total 1x20=20 points, John's new score is 61 points
John's round 2, choose a number from 1 to 20 inclusive, or 25: 19
It's a double, total 2x19=38 points, John's new score is 23 points
John's round 3, choose a number from 1 to 20 inclusive, or 25: 7
It's a triple, total 3x7=21 points, John's new score is 2 points
Next player...
Tom's turn 5, current score is 161 points
Tom's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a triple, total 3x20=60 points, Tom's new score is 101 points
Tom's round 2, choose a number from 1 to 20 inclusive, or 25: 20
It's a triple, total 3x20=60 points, Tom's new score is 41 points
Tom's round 3, choose a number from 1 to 20 inclusive, or 25: 20
It's a double, total 2x20=40 points, Tom is busted, reverting to
initial score 161! Try again next turn
Next player...
Alan's turn 5, current score is 26 points
Alan's round 1, choose a number from 1 to 20 inclusive, or 25: 20
It's a double, total 2x20=40 points, Alan is busted, reverting to
initial score 26! Try again next turn
Next player...
John's turn 5, current score is 2 points
John's round 1, choose a number from 1 to 20 inclusive, or 25: 1
It's a double, total 2x1=2 points, John wins in 5 turns!
Tom's score is 161
Alan's score is 26
John's score is 0
```

(15 marks)

Question 4 (35 marks)

This question covers materials up to seminar 4. Employ structure programming and use of functions to make the program modular.

Silver Screen is a cinema operator looking for a simple application to help them manage seat bookings.

- A cinema has between 3 to 10 (both inclusive) rows of seats. Each row has the same number of seats which can range between 3 to 10 (both inclusive).
- Rows are numbered ‘01’, ‘02’, ‘03’, …, starting from the row just in front of the screen. Seats are numbered ‘A’, ‘B’, ‘C’, …, starting from the left when facing the screen.
- When displaying a seating plan, available seats are marked with ‘O’, while booked seats are marked with ‘X’. An example of a 5 rows x 8 seats cinema with 4 seats booked is shown below.

```
                Screen

   A B C D E F G H
01 O O O O O O O O
02 O O O O O O O O
03 O O O O O O O O
04 O O O O O O O O
05 O O X X X X O O
```

(a) Write a menu-driven application to manage the seat booking process. The application will first ask the user to input the number of rows and the number of seats per row to initialize a seating plan as a nested list. This nested list shall consist of a list of rows where each row is a list of seats. For example, the initial seating plan of the above cinema is represented by the nested list below.

```
seatingPlan = [['O', 'O', 'O', 'O', 'O', 'O', 'O', 'O'],
               ['O', 'O', 'O', 'O', 'O', 'O', 'O', 'O'],
               ['O', 'O', 'O', 'O', 'O', 'O', 'O', 'O'],
               ['O', 'O', 'O', 'O', 'O', 'O', 'O', 'O'],
               ['O', 'O', 'O', 'O', 'O', 'O', 'O', 'O']]
```

- Next, the application will go into a loop, presenting the user with a menu of options. The user should be able to choose between displaying the current seating plan, booking of seats, or quitting the application. Structure your program using functions to make it modular and easy to add new options in the future.

- If the user chooses to display the seating plan, show the seating plan in a format similar to the above and return to the main menu.

- If the user chooses to book seats, display the current seating plan and ask the user to input the location and the number of seats to book in the format '@NN-#', where **@** represents the seat label A,B,C,…, **NN** is a 2-digit number representing the row, and **#** is the number of seats to book. For example, 'C05-4' is a request to book 4 seats starting from row 5 seat C. Users can only book continuous seats on the same row. If all seats are available, the function will update the booked seats with 'X' and display the updated seating plan. If not, it will display an error message and return to main menu.

- Allow the user to continue choosing other menu options until he/she chooses option to quit.

- Sample runs are given below. Submit and paste screenshots to show at least FOUR different features of your new application.

<u>Sample 1:</u>

```
Enter number of rows (3-10 or 0 to quit): 5
Enter number of seats per row (3-10 or 0 to quit): 11
Invalid input! Please try again
Enter number of seats per row (3-10 or 0 to quit): 0
Program end
```

<u>Sample 2:</u>

```
Enter number of rows (3-10 or 0 to quit): 6
Enter number of seats per row (3-10 or 0 to quit): 8

Booking Menu
------------
1. Show seats
2. Book seats
0. Quit
Enter option: 1

       Screen

   A B C D E F G H
01 O O O O O O O O
02 O O O O O O O O
03 O O O O O O O O
04 O O O O O O O O
05 O O O O O O O O
06 O O O O O O O O


Booking Menu
------------
1. Show seats
2. Book seats
0. Quit
Enter option: 2

       Screen

   A B C D E F G H
01 O O O O O O O O
02 O O O O O O O O
03 O O O O O O O O
04 O O O O O O O O
05 O O O O O O O O
06 O O O O O O O O

Enter seats to book (eg C05-4) or 0 to cancel): ABC
Invalid input! Please enter in @NN-# format where @:seat label,
NN:row, #:number of seats
Enter seats to book (eg C05-4) or 0 to cancel): A07-1
Invalid row! Please enter in @NN-# format where @:seat label, NN:row,
#:number of seats
Enter seats to book (eg C05-4) or 0 to cancel): I01-1
Invalid seat label! Please enter in @NN-# format where @:seat label,
NN:row, #:number of seats
Enter seats to book (eg C05-4) or 0 to cancel): D03-3
Seats booked
```

```
       Screen

   A B C D E F G H
01 O O O O O O O O
02 O O O O O O O O
03 O O O X X X O O
04 O O O O O O O O
05 O O O O O O O O
06 O O O O O O O O


Booking Menu
------------
1. Show seats
2. Book seats
0. Quit
Enter option: 2

       Screen

   A B C D E F G H
01 O O O O O O O O
02 O O O O O O O O
03 O O O X X X O O
04 O O O O O O O O
05 O O O O O O O O
06 O O O O O O O O

Enter seats to book (eg C05-4) or 0 to cancel): A03-4
Not all seats at those locations are available! Please try again
Enter seats to book (eg C05-4) or 0 to cancel): A03-3
Seats booked

       Screen

   A B C D E F G H
01 O O O O O O O O
02 O O O O O O O O
03 X X X X X X O O
04 O O O O O O O O
05 O O O O O O O O
06 O O O O O O O O


Booking Menu
------------
1. Show seats
2. Book seats
0. Quit
Enter option: 2
```

```
       Screen

   A B C D E F G H
01 O O O O O O O O
02 O O O O O O O O
03 X X X X X X O O
04 O O O O O O O O
05 O O O O O O O O
06 O O O O O O O O

Enter seats to book (eg C05-4) or 0 to cancel): A01-9
Invalid number of seats! Please enter in @NN-# format where @:seat
label, NN:row, #:number of seats
Enter seats to book (eg C05-4) or 0 to cancel): 0
Cancelled

Booking Menu
------------
1. Show seats
2. Book seats
0. Quit
Enter option: 0
Program end
```

(20 marks)

(b) Enhance the above application with a new option to auto select the best seats to book. With this option, the user will only need to input the number of seats required and the application will find the best seats to book. It will start from the last row and work forward towards the screen. In each row, it will try to find available seats as close to the centre of the row as possible. Selected seats should form a continuous row that has the required number of seats. The seats cannot span multiple rows and cannot have gaps in between seats.

- If such seats are found, the application will ask the user for confirmation to book. If the user agrees, update the seating plan marking those seats as 'X' and display the updated seating plan.
- If the user is not satisfied with the seats found, continue to find the next best seats until he/she agrees or no more available seats can be found.
- The user can opt to return to main menu at any step.
- Sample runs are given below. Submit and paste screenshots to show at least FOUR key features of your new application.

```
Enter number of rows (3-10 or 0 to quit): 4
Enter number of seats per row (3-10 or 0 to quit): 6

Booking Menu
------------
1. Show seats
```

```
2. Book seats
3. Auto select seats
0. Quit
Enter option: 3
Enter number of seats (0 to cancel): 7
Invalid input! Please try again
Enter number of seats (0 to cancel): 0
Cancelled

Booking Menu
------------
1. Show seats
2. Book seats
3. Auto select seats
0. Quit
Enter option: 3
Enter number of seats (0 to cancel): 1

     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O O O O O O
04 O O O O O O

Would you like to book the seat C04 (y/n or 0 to cancel): n

     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O O O O O O
04 O O O O O O

Would you like to book the seat D04 (y/n or 0 to cancel): n

     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O O O O O O
04 O O O O O O

Would you like to book the seat B04 (y/n or 0 to cancel): n
```

```
     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O O O O O O
04 O O O O O O

Would you like to book the seat E04 (y/n or 0 to cancel): y
Seats booked

     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O O O O O O
04 O O O O X O


Booking Menu
------------
1. Show seats
2. Book seats
3. Auto select seats
0. Quit
Enter option: 3
Enter number of seats (0 to cancel): 5

     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O O O O O O
04 O O O O X O

Would you like to book the seats A03 to E03 (y/n or 0 to cancel): n

     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O O O O O O
04 O O O O X O

Would you like to book the seats B03 to F03 (y/n or 0 to cancel): y
Seats booked
```

```
     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O X X X X X
04 O O O O X O


Booking Menu
------------
1. Show seats
2. Book seats
3. Auto select seats
0. Quit
Enter option: 3
Enter number of seats (0 to cancel): 6

     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O X X X X X
04 O O O O X O

Would you like to book the seats A02 to F02 (y/n or 0 to cancel): n

     Screen

   A B C D E F
01 O O O O O O
02 O O O O O O
03 O X X X X X
04 O O O O X O

Would you like to book the seats A01 to F01 (y/n or 0 to cancel): n
Unable to find available seats

Booking Menu
------------
1. Show seats
2. Book seats
3. Auto select seats
0. Quit
Enter option: 0
Program end
```

(15 marks)

---- END OF ASSIGNMENT ----